ant neuro
inspiring technology

McHacks

NOVA

UBEC

Buildathon 2026

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NOVA BIOTECH MCGILL

NOVA is a student-run Biotech and Entrepreneurship organization operating under McGill's Faculty of Science. We aim to bridge the intersection between business and science to empower the next generation of biotech startups.

Our mission is based on 3 pillars:

Education

Host speaker events with leading voices in biotech, entrepreneurship, and venture capital.

Internship Pipeline

Connect a large talent pool of passionate students to our vast network of industry partners.

Buildathon

Host buildathons to encourage hands-on innovation and real-world problem solving

Our team is composed of 20 students with a diverse range of academic backgrounds, all united by their passion for biotechnology

Join us for McHacks 14
January 16–17, 2027

McHacks

McHacks is McGill's largest annual hackathon, now in its 14th year. Over an intense 24 hours, **750+ students** from across North America team up to turn bold ideas into working products, backed by mentors, workshops, and a community that celebrates builders of every background and skill level. Our goal is simple: give the next generation of builders the tools, network, and momentum to **turn ambitious ideas into real-world impact**.

We're excited to bring that same builder energy to this buildathon, where science and entrepreneurship meet. Whether this is your first time building something from scratch or your tenth, welcome to the table.

mchacks.ca
@mcgillhacks
contact@mchacks.ca
/company/mchacks/

CUBEC

CUBEC is a student-led, national, government-registered non-profit promoting **education**, **inclusion**, and **awareness** in the biomedical engineering industry and research community among undergraduates across Canada.

Our mission:

- Provide a platform for students to network and build meaningful connections.
- Expose and inform undergraduate biomedical engineering students across Canada about internship and post-graduate options.

Throughout the year, our initiatives consist of community-oriented events, youth BME outreach, and online content to help guide undergraduates through their paths in BME. We host an annual conference, bringing together over 1,000 attendees across four Canadian host cities. Additionally, we host the True North Biomedical Engineering Competition, one of Canada’s first BME-focused design competitions. We are excited to be working with NOVA and McHacks at this year’s Buildathon!

cubec.info
cubec.nfpo
CUBEC

ANT Neuro

ANT Neuro is devoted to providing cutting-edge solutions for neuroscience and neurodiagnostics, and particularly aiming at systems and products which inspire scientists and clinicians to undertake new actions and gain better insight into the human brain.

Special thank you to Johnathan Drucker, Director of BD at ANT Neuro, for his support in making this challenge possible!

As this years sponsor, they have kindly provided us with both EEGs and FNIRS hardware for participants to use in their project on the build day (September 12th).

To read more about their products and solutions, please visit ant-neuro.com

@ant_neuro

ANT Neuro

Buildathon Timeline

Virtual dates

August 12

- Challenge begins
- Primary dataset and resources provided
- Discord server is active with mentors.

August 13 - August 27

- Period for research, brainstorming, and code experimentation

August 28 - September 12

- Refining final code and presentations

In-person dates

September 12

- Opening ceremonies
- Access to hardware (EEG)
- Build time to implement solutions
- Key Note Speakers + Neetworking

September 13

- All deliverables due
- Presentations and judging
- Closing ceremonies

*Detailed schedules will be released closer to the date

THE CHALLENGE!

NeuroLoop

This years challenge will give participants the chance to work with *industry grade* **EEG** technology from ANT Neuro.

The Task

Teams will build a closed-loop system that utilizes live EEG and fNIRS data to sense a persons mental state to assess cognitive or physical performance. In response, the system will adapt its output accordingly, whether the output be messages to user or changing a virtual environment or task.

An example could be reading a basketball players brain during practice, and providing adaptive feedback and changing training exercises via prompts to maximize performance. Or a personalized study tool connected to an alarm, or other hardware.

Core Requirements

- Build an algorithm that can recognize meaningful signals or indicators within a 12 lead EEG data set (Feature Extraction)
- Correlate these signals to real time performance to predict cognitive or physical success at a task
- Create a form of output to communicate with user. This could range from a simple feedback pop-up message to an entire virtual environment or physical
- Have real world implication and uses, with a desired target market and projected business strategy

Resources

- EEG Data sets from ANT neuro
- Open Source EEG data to solidify and build your algorithm
- Code templates and resources
- Mentorship through Discord chat & call, including from ANT Neuro directly
- Access to EEG hardware during the buildathon weekend
- Comprehensive guides and tutorials on EEG/fNIRS technology

NeuroLoop

Deliverables for September 13th

- Code file with adequate docstrings explaining the basis of the algorithm you created. Must include team number and member names at the top. Acceptable formats include: **.py, .java, .cpp**

- Pitch presentation deck in both **.pptx and .pdf** format, with file naming convention **"teamnumber_buildathon"**. Presentation should include the following sections in any order:
  - Explanation of ANTneuro's EEGgo and how it works
  - General overview of algorithm created and how it was built
  - Difficulties faced (if any) and how the team overcame them
  - Chosen application of technology and its benefits (what real world problem or need does your product adress?)
  - Demo
  - Future directions of technical features
  - Business plan and projected revenue streams

Remember, this is a pitch! Although you should cover all aforementioned topics, do so in a creative and engaging manner to appeal to investors, there is no "right" structure.

AI Policy

We recommend minimizing the use of AI for your own learning and benefit, however, we understand that it may be a helpful tool throughout the project so long as its utilized in a smart manner.

If you utilize AI, you must present how exactly you used it, and demonstrate an understanding of any steps or lines of code it was involved in.

You may be asked to elaborate on the spot to ensure understanding.

Judging Criteria

Presentation skills, structure, flow, and visual appeal

Technical understanding and implementation, including algorithm accuracy, features, and output

Chosen application's benefits and impact

Comprehensive buisness plan, realistic goals, and projections

*Not in order of importance

Resources

VERY IMPORTANT!

We will be using this Discord channel for all **future announcements and schedules**. It will also be the main source of mentorship, resources, and a platform to ask any questions, logistical and technical.

Make sure you check it regularly!

Join the NOVA Buildathon Discord Server!

Check out the NOVA Buildathon community on Discord - hang out with 5 other members and enjoy free voice and text chat.

Discord

Link also provided in email*

Data files will be available on here as well

Best of luck!

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